

M12x Unit Catalogue

Graphs, mechanisms, tables, and reference hypotheses

Expanded Milestone 1 Part 2 (U1–U7)

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Compiled from locked mechanism cards (2026-07-16)

Abstract

Quick-reference catalogue of the seven accepted M12x units. Each unit is a graph G together with a deterministic mechanism assignment on every non-source. Cells are (unit, $t \in N$, config $\in \{\text{ECAI}, \text{AAMAS}\}$); this document records inputs only (not learner outputs). Units with several targets list each target separately, with its own background-knowledge predictors, labels $E^\pm$, and reference hypothesis $\mathcal{H}_t^*$.

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1 Shared regime

Alphabet

$K = \{0, 1, 2\}$ for every variable.

DGP

Full source factorial over $K^{|S|}$, then propagate mechanisms in topological order. Mechanisms are drawn from $\{\text{copy}, \text{min}, \text{max}\}$, with copy only when $|\text{Pa}(y)| = 1$.

Labels (per target t)

$E^+ = \{t \neq 0\}$, $E^- = \{t = 0\}$ (1-based sample ids on $\mathcal{D}$).

Background knowledge

Every column except t : exact-value predicates *_val_0/1/2 plus definitional nonzero facts

```
x_nz(A) :- x_val_1(A).
x_nz(A) :- x_val_2(A).
```

Descendants of t remain in BK as distractors; citing them in a learned rule is an explicit failure mode.

Reference hypothesis

Compact **nz**-language rule preferred; equivalent **val**-expansions are extensionally acceptable but inferior.

Cell count

7 units, 11 (unit, target) pairs, 22 cells after $\times \{\text{ECAI}, \text{AAMAS}\}$.

At-a-glance

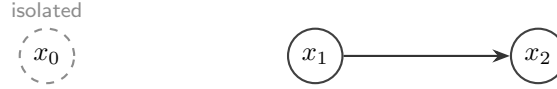
ID	Graph	Mechanisms	Targets N	$ S $	Rows	Cells
U1	separator (+ isolated x_0)	$x_2 := \text{copy}(x_1)$	$\{x_2\}$	2	9	2
U2	collider	$x_2 := \min(x_0, x_1)$	$\{x_2\}$	2	9	2
U3	collider	$x_2 := \max(x_0, x_1)$	$\{x_2\}$	2	9	2
U4	fork	$x_1 := \text{copy}(x_0), x_2 := \text{copy}(x_0)$	$\{x_1, x_2\}$	1	3	4
U5	chain	$x_1 := \text{copy}(x_0), x_2 := \text{copy}(x_1)$	$\{x_1, x_2\}$	1	3	4
U6	Fabrizio G1	$x_2 := \min, x_3 := \min$	$\{x_2, x_3\}$	2	9	4
U7	Fabrizio G1	$x_2 := \max, x_3 := \max$	$\{x_2, x_3\}$	2	9	4
11 targets						22

2 U1 — Separator + copy

Fixture key `m12.u1.separator.copy`
Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$
Learning targets $\{x_2\}$ (single target)

Lesson. Unique parent / copy with an isolated distractor.

2.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_1 \rightarrow x_2\}, \quad S = \{x_0, x_1\}, \quad N = \{x_2\}.$$

2.2 Mechanisms

$$x_2 := \text{copy}(x_1).$$

Boolean nonzero sketch: $\Phi_{x_2} = z_1$.

2.3 Table $\mathcal{D}$

Full factorial over (x_0, x_1) ; then $x_2 = x_1$.

id	x_0	x_1	$x_2 = \text{copy}(x_1)$
1	0	0	0
2	0	1	1
3	0	2	2
4	1	0	0
5	1	1	1
6	1	2	2
7	2	0	0
8	2	1	1
9	2	2	2

2.4 Target $t = x_2$

Single target — sink

Ancestors $\{x_1\}$; distractor $\{x_0\}$ (isolated, in BK); descendants $\emptyset$.
BK predictors: x_0, x_1 (exclude only x_2).
 $E^+ = \{2, 3, 5, 6, 8, 9\}$ (6 rows); $E^- = \{1, 4, 7\}$ (3 rows).

Reference $\mathcal{H}_{x_2}^*$:

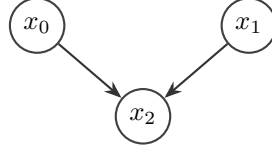
$$\text{x2(A)} \text{ :- x1.nz(A) .}$$

3 U2 — Collider + min

Fixture key `m12.u2.collider.min`
Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$
Learning targets $\{x_2\}$ (single target)

Lesson. Conjunctive / AND nonzero rule.

3.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_0 \rightarrow x_2, x_1 \rightarrow x_2\}, \quad S = \{x_0, x_1\}, \quad N = \{x_2\}.$$

3.2 Mechanisms

$$x_2 := \min(x_0, x_1).$$

Boolean nonzero sketch: $\Phi_{x_2} = z_0 \wedge z_1$.

3.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \min$
1	0	0	0
2	0	1	0
3	0	2	0
4	1	0	0
5	1	1	1
6	1	2	1
7	2	0	0
8	2	1	1
9	2	2	2

3.4 Target $t = x_2$

Single target — sink

Ancestors $\{x_0, x_1\}$; other distractors $\emptyset$; descendants $\emptyset$.
 BK predictors: x_0, x_1 .
 $E^+ = \{5, 6, 8, 9\}$ (4 rows); $E^- = \{1, 2, 3, 4, 7\}$ (5 rows).
 On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$ and $x_1 \neq 0$.

Reference $\mathcal{H}_{x_2}^*$:

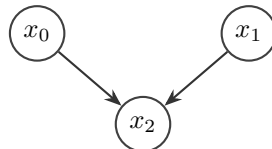
$$\text{x2(A)} \text{ :- } \text{x0_nz(A)}, \text{ x1_nz(A)}.$$

4 U3 — Collider + max

Fixture key `m12_u3_collider_max`
 Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$
 Learning targets $\{x_2\}$ (single target)

Lesson. Disjunctive / OR nonzero rule.
Same edge set as U2; mechanism differs (max vs min).

4.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_0 \rightarrow x_2, x_1 \rightarrow x_2\}, \quad S = \{x_0, x_1\}, \quad N = \{x_2\}.$$

4.2 Mechanisms

$$x_2 := \max(x_0, x_1).$$

Boolean nonzero sketch: $\Phi_{x_2} = z_0 \vee z_1$.

4.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \max$
1	0	0	0
2	0	1	1
3	0	2	2
4	1	0	1
5	1	1	1
6	1	2	2
7	2	0	2
8	2	1	2
9	2	2	2

4.4 Target $t = x_2$

Single target — sink

Ancestors $\{x_0, x_1\}$; other distractors $\emptyset$; descendants $\emptyset$.

BK predictors: x_0, x_1 .

$E^+ = \{2, \dots, 9\}$ (8 rows); $E^- = \{1\}$ (1 row).

On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$ or $x_1 \neq 0$.

Reference $\mathcal{H}_{x_2}^*$:

$x_2(A) :- x_0.\text{nz}(A).$

$x_2(A) :- x_1.\text{nz}(A).$

5 U4 — Fork + double copy

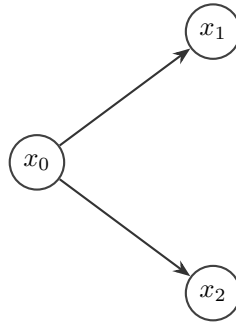
Fixture key `m12_u4_fork_double_copy`

Sources / rows $|S| = 1 \Rightarrow |\mathcal{D}| = 3$

Learning targets $\{x_1, x_2\}$ — two targets (symmetric children)

Lesson. Sibling distractor under perfect extensional tie to the parent.

5.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_0 \rightarrow x_1, x_0 \rightarrow x_2\}, \quad S = \{x_0\}, \quad N = \{x_1, x_2\}.$$

5.2 Mechanisms

$$\begin{aligned}x_1 &:= \text{copy}(x_0), \\x_2 &:= \text{copy}(x_0).\end{aligned}$$

Boolean nonzero sketch: $\Phi_{x_1} = \Phi_{x_2} = z_0$. On $\mathcal{D}$, $x_0 = x_1 = x_2$ in every row.

5.3 Table $\mathcal{D}$

id	x_0	$x_1 = \text{copy}(x_0)$	$x_2 = \text{copy}(x_0)$
1	0	0	0
2	1	1	1
3	2	2	2

5.4 Target $t = x_1$

Target 1 of 2 — child of x_0

Ancestors $\{x_0\}$; sibling distractor $\{x_2\}$ (**in BK**); descendants $\emptyset$.
 BK predictors: x_0, x_2 (exclude only x_1).
 $E^+ = \{2, 3\}$; $E^- = \{1\}$.
 On $\mathcal{D}$: $x_1 \neq 0$ iff $x_0 \neq 0$ iff $x_2 \neq 0$.

Reference $\mathcal{H}_{x_1}^*$:

$$\text{x1(A)} \text{ :- x0.nz(A) .}$$

Do not cite sibling x_2 (even though extensionally equivalent).

5.5 Target $t = x_2$

Target 2 of 2 — child of x_0 (symmetric)

Ancestors $\{x_0\}$; sibling distractor $\{x_1\}$ (**in BK**); descendants $\emptyset$.
 BK predictors: x_0, x_1 (exclude only x_2).
 $E^+ = \{2, 3\}$; $E^- = \{1\}$.

Reference $\mathcal{H}_{x_2}^*$:

$$\text{x2(A)} \text{ :- x0.nz(A) .}$$

Do not cite sibling x_1 .

6 U5 — Chain + double copy

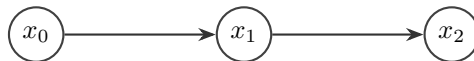
Fixture key `m12_u5_chain_double_copy`

Sources / rows $|S| = 1 \Rightarrow |\mathcal{D}| = 3$

Learning targets $\{x_1, x_2\}$ — two targets (intermediate + sink)

Lesson. Parent vs ancestor cut; when learning the intermediate, the descendant stays in BK.

6.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_0 \rightarrow x_1, x_1 \rightarrow x_2\}, \quad S = \{x_0\}, \quad N = \{x_1, x_2\}.$$

6.2 Mechanisms

$$\begin{aligned}x_1 &:= \text{copy}(x_0), \\x_2 &:= \text{copy}(x_1).\end{aligned}$$

Boolean nonzero sketch: $\Phi_{x_1} = \Phi_{x_2} = z_0$. On $\mathcal{D}$, $x_0 = x_1 = x_2$ in every row.

6.3 Table $\mathcal{D}$

id	x_0	$x_1 = \text{copy}(x_0)$	$x_2 = \text{copy}(x_1)$
1	0	0	0
2	1	1	1
3	2	2	2

6.4 Target $t = x_1$ (intermediate)

Target 1 of 2 — intermediate; descendant distractor cell

Ancestors $\{x_0\}$; other distractors $\emptyset$; descendants $\{x_2\}$ (**in BK**; citing = failure mode).
 BK predictors: x_0, x_2 (exclude only x_1).
 $E^+ = \{2, 3\}$; $E^- = \{1\}$.
 On $\mathcal{D}$: $x_1 \neq 0$ iff $x_0 \neq 0$ iff $x_2 \neq 0$.

Reference $\mathcal{H}_{x_1}^*$:

$$\text{x1(A)} \text{ :- x0_nz(A) .}$$

Must not cite descendant x_2 .

6.5 Target $t = x_2$ (sink)

Target 2 of 2 — sink; parent vs ancestor

Ancestors $\{x_0, x_1\}$ (parent x_1 preferred; x_0 is grandparent); descendants $\emptyset$.
 BK predictors: x_0, x_1 (exclude only x_2).
 $E^+ = \{2, 3\}$; $E^- = \{1\}$.

Reference $\mathcal{H}_{x_2}^*$ (direct parent):

$$\text{x2(A)} \text{ :- x1_nz(A) .}$$

Note: $\text{x2(A)} \text{ :- x0_nz(A) .}$ is extensionally correct on $\mathcal{D}$ but is a parent-vs-ancestor divergence, not the reference.

7 U6 — Fabrizio G1 + AND cone

Fixture key `m12_u6_g1_and_cone`

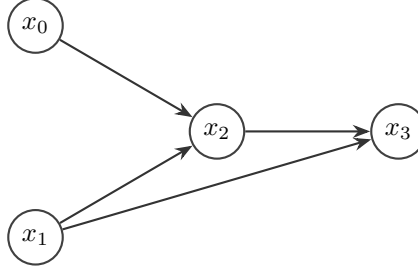
Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$

Learning targets $\{x_2, x_3\}$ — **two targets** (intermediate + sink)

Lesson. Four-node G1; two-level AND; descendant distractor when learning x_2 .

Graph provenance: Fabrizio G1 (ArgCausalDisco four-node example). Mechanisms/DGP are project-chosen.

7.1 Graph



$V = \{x_0, x_1, x_2, x_3\}$, $E = \{x_0 \rightarrow x_2, x_1 \rightarrow x_2, x_1 \rightarrow x_3, x_2 \rightarrow x_3\}$,
 $S = \{x_0, x_1\}$, $N = \{x_2, x_3\}$, $\text{Pa}(x_2) = \{x_0, x_1\}$, $\text{Pa}(x_3) = \{x_1, x_2\}$.

7.2 Mechanisms

$$\begin{aligned} x_2 &:= \min(x_0, x_1), \\ x_3 &:= \min(x_1, x_2). \end{aligned}$$

Boolean nonzero sketch: $\Phi_{x_2} = \Phi_{x_3} = z_0 \wedge z_1$.

Composition identity on $\mathcal{D}$: $\min(x_1, \min(x_0, x_1)) = \min(x_0, x_1)$, so $x_3 = x_2$ in every row.

7.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \min(x_0, x_1)$	$x_3 = \min(x_1, x_2)$
1	0	0	0	0
2	0	1	0	0
3	0	2	0	0
4	1	0	0	0
5	1	1	1	1
6	1	2	1	1
7	2	0	0	0
8	2	1	1	1
9	2	2	2	2

7.4 Target $t = x_2$ (intermediate)

Target 1 of 2 — intermediate; descendant x_3 in BK

Ancestors $\{x_0, x_1\}$; descendants $\{x_3\}$ (in BK; citing = failure mode).

BK predictors: x_0, x_1, x_3 (exclude only x_2).

$E^+ = \{5, 6, 8, 9\}$ (4); $E^- = \{1, 2, 3, 4, 7\}$ (5).

On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$ and $x_1 \neq 0$ (also iff $x_3 \neq 0$).

Reference $\mathcal{H}_{x_2}^*$:

$$\text{x2(A)} \text{ :- } \text{x0_nz(A)}, \text{ x1_nz(A)}.$$

Must not cite descendant x_3 (despite $x_3 = x_2$ on $\mathcal{D}$).

7.5 Target $t = x_3$ (sink)

Target 2 of 2 — sink

Ancestors $\{x_0, x_1, x_2\}$; parents $\{x_1, x_2\}$; descendants $\emptyset$.

BK predictors: x_0, x_1, x_2 (exclude only x_3).

$E^+ = \{5, 6, 8, 9\}$ (4); $E^- = \{1, 2, 3, 4, 7\}$ (5).

On $\mathcal{D}$: $x_3 \neq 0$ iff $x_1 \neq 0$ and $x_2 \neq 0$ (equivalently iff $x_0 \neq 0$ and $x_1 \neq 0$).

Reference $\mathcal{H}_{x_3}^*$ (parent-based):

$$x_3(A) \text{ :- } x_1.\text{nz}(A), x_2.\text{nz}(A).$$

Alternative (extensionally equivalent on $\mathcal{D}$): $x_3(A) \text{ :- } x_0.\text{nz}(A), x_1.\text{nz}(A)$. — record if recovered; prefer parent form as $\mathcal{H}^$.*

8 U7 — Fabrizio G1 + OR cone

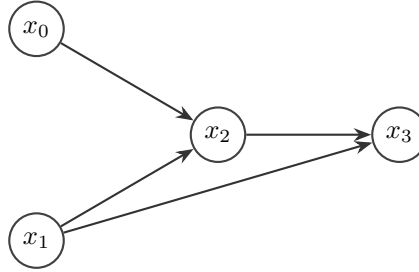
Fixture key `m12_u7_g1_or_cone`

Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$

Learning targets $\{x_2, x_3\}$ — **two targets** (intermediate + sink)

Lesson. Same G1 as U6; two-level OR; descendant distractor when learning x_2 .
Same edge set as U6; mechanism differs (max vs min).

8.1 Graph



$$V = \{x_0, x_1, x_2, x_3\}, \quad E = \{x_0 \rightarrow x_2, x_1 \rightarrow x_2, x_1 \rightarrow x_3, x_2 \rightarrow x_3\},$$

$$S = \{x_0, x_1\}, \quad N = \{x_2, x_3\}.$$

8.2 Mechanisms

$$x_2 := \max(x_0, x_1),$$

$$x_3 := \max(x_1, x_2).$$

Boolean nonzero sketch: $\Phi_{x_2} = \Phi_{x_3} = z_0 \vee z_1$.

Composition identity on $\mathcal{D}$: $\max(x_1, \max(x_0, x_1)) = \max(x_0, x_1)$, so $x_3 = x_2$ in every row.

8.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \max(x_0, x_1)$	$x_3 = \max(x_1, x_2)$
1	0	0	0	0
2	0	1	1	1
3	0	2	2	2
4	1	0	1	1
5	1	1	1	1
6	1	2	2	2
7	2	0	2	2
8	2	1	2	2
9	2	2	2	2

8.4 Target $t = x_2$ (intermediate)

Target 1 of 2 — intermediate; descendant x_3 in BK

Ancestors $\{x_0, x_1\}$; descendants $\{x_3\}$ (in BK; citing = failure mode).

BK predictors: x_0, x_1, x_3 (exclude only x_2).

$E^+ = \{2, \dots, 9\}$ (8); $E^- = \{1\}$ (1).

On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$ or $x_1 \neq 0$ (also iff $x_3 \neq 0$).

Reference $\mathcal{H}_{x_2}^*$:

`x2(A) :- x0.nz(A).`

`x2(A) :- x1.nz(A).`

Must not cite descendant x_3 .

8.5 Target $t = x_3$ (sink)

Target 2 of 2 — sink

Ancestors $\{x_0, x_1, x_2\}$; parents $\{x_1, x_2\}$; descendants $\emptyset$.

BK predictors: x_0, x_1, x_2 (exclude only x_3).

$E^+ = \{2, \dots, 9\}$ (8); $E^- = \{1\}$ (1).

On $\mathcal{D}$: $x_3 \neq 0$ iff $x_1 \neq 0$ or $x_2 \neq 0$ (equivalently iff $x_0 \neq 0$ or $x_1 \neq 0$).

Reference $\mathcal{H}_{x_3}^*$ (parent-based):

`x3(A) :- x1.nz(A).`

`x3(A) :- x2.nz(A).`

Alternative (extensionally equivalent on $\mathcal{D}$):

`x3(A) :- x0.nz(A).`

`x3(A) :- x1.nz(A).`

Source pointers

Mechanism cards

`docs/research/milestone_plans/milestone1_part2/mechanism_cards/`

Unit set

`milestone1_part2_expanded_unit_set.md`

Approach

`milestone1_part2_expanded_approach.md`

Implementation

`causal/experiments/handcrafted_m12x.py`